

FINAL EXAMINATION

Course: CS486 – Introduction to Database System

Time: 10 minutes

Term: III – Academic year: 2020-2021

Lecturer: Dr. Le Thi Nhan

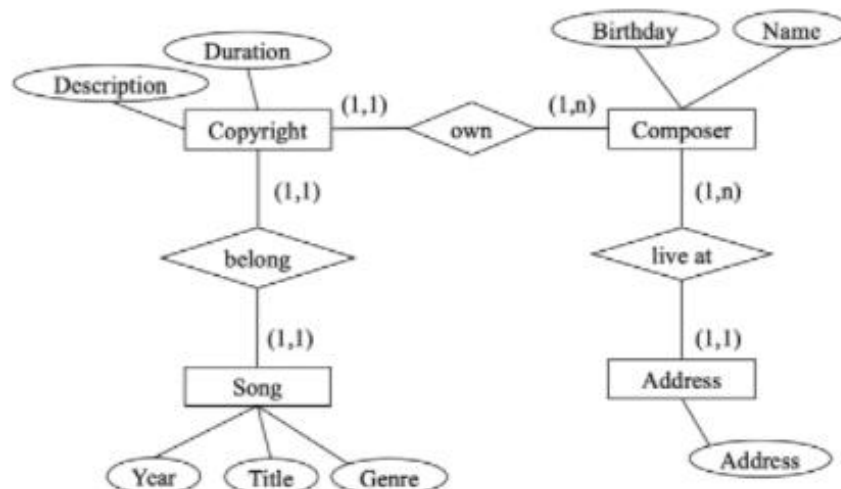
Student name:

Student ID:

(Notes: Laptops and mobile devices are allowed)

Question 1:

What design rules are violated in the figure below? What modifications would you suggest?



Question 2:

Given the relation Song as the right hand side. How many operations of Relational Algebra are required to find titles of the genre “Rock” which have *exactly* 2 different release years? What are these operations?

Title	Year	Genre
Hotel California	1976	Rock
Hotel California	2002	Acoustic
Heal the world	1991	Pop
Don't cry	1991	Rock
Don't cry	2009	Rock

Question 3:

Why does the Tableau algorithm guarantee that the natural join must be lossless when the resulting matrix contains a row with all a values?

Question 4:

Consider the relation $R(A)$ with two tuples $\{(1), (2)\}$.

Suppose the transaction $T1$ and $T2$ want to update the attribute A as follows,

$T1$: update R set $A=A+2$

$T2$: update R set $A=A*2$

What are the possible final states of R if the two transactions are submitted under the isolation and atomicity properties?

Question 5:

Does the final database state correspond to the effect of a sequence of transactions during recovery of redo/undo method? Support your answer.

--The end--